

ANISSA ZUHRITA

Environmental GIS & Remote Sensing Specialist | Climate & Environmental Analyst

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PROFESSIONAL SUMMARY

Environmental GIS and Remote Sensing Specialist with professional experience supporting 40+ environmental and development projects through GIS analysis, environmental assessment, spatial data integration, field surveys, and technical documentation. Experienced in AMDAL, UKL-UPL, DELH, environmental baseline assessment, and spatial analysis across infrastructure, plantation, housing, and oil & gas projects.

Holds an MSc in Remote Sensing from Universitas Gadjah Mada with research experience spanning carbon and biomass assessment, climate variability, hydrological modelling, and satellite-based environmental monitoring. Experienced in integrating satellite imagery, field observations, environmental datasets, and climate records into quantitative and spatial analyses using Google Earth Engine, QGIS, ArcGIS, and Python.

Brings additional experience in technical communication and capacity building, including invited speaking at international online training and presenting environmental and remote sensing research in academic forums.

PROFESSIONAL EXPERIENCE

GIS & Environmental Impact Analyst

November 2021 – January 2024

PT Anugrah Agung Nusantara — Jambi, Indonesia

- Supported 40+ environmental and development projects across infrastructure, plantation, housing, and oil & gas sectors through GIS analysis, environmental assessment, field data collection, and technical documentation.
- Supported the preparation of AMDAL, UKL-UPL, DELH, and technical environmental studies, integrating spatial data and environmental information into project assessments.
- Served as GIS/mapping specialist for the AMDAL of a planned ± 77.5 km coal transportation road across Muaro Jambi and Batang Hari, integrating peatland, PIPPIB, land cover, topography, hydrology, and spatial-planning datasets through spatial overlay analysis.
- Prepared environmental baseline and thematic maps covering administrative boundaries, land use/land cover, hydrology, infrastructure, environmental constraints, and sampling locations.
- Conducted spatial analyses including overlay analysis, watershed delineation, proximity analysis, land suitability assessment, and environmental sensitivity mapping to support project planning and environmental impact assessment.
- Supported environmental studies for oil & gas projects through spatial analysis of spatial planning, forest areas, geology, hydrology, and surrounding environmental conditions.
- Supported 15+ residential UKL-UPL projects, including runoff estimation, drainage analysis, and spatial planning of environmental management measures.
- Contributed to a technical feasibility study for palm oil mill effluent land application (LCPKS) by integrating topography, soil characteristics, and geophysical survey information into a spatial analysis framework.
- Conducted field surveys and GPS-based spatial data collection to support environmental baseline assessment, data verification, and ground-truthing.

SELECTED CLIMATE & ENVIRONMENTAL RESEARCH

Influence of ENSO on Rainfall Variability in Central Sulawesi

Co-Author / Climate Data Analyst · Buletin GAW Bariri, Vol. 6 No. 2, 2025

- Analysed relationships between Oceanic Niño Index (ONI) and rainfall anomalies across four BMKG meteorological stations using Pearson correlation, R^2 , and statistical significance testing over the 1991–2024 period.
- Evaluated the influence of different El Niño and La Niña intensities and found that ENSO impacts vary spatially between locations, highlighting the importance of regional climate characteristics in climate-related planning.

Flood Potential Assessment of the Way Urang Sub-Watershed

Researcher / Co-Author · International Journal for Disaster and Development Interface, Vol. 5 Issue 2, 2025

- Integrated Sentinel-2, DEMNAS, rainfall records, vegetation observations, soil infiltration measurements, and river morphometry to assess flood potential.

- Developed spatial runoff-coefficient mapping using the Cook Method and estimated peak discharge using the Rational Method.
- Compared estimated peak discharge with field-derived channel capacity to support spatially targeted flood-risk interpretation and mitigation planning.

Mangrove Biomass Assessment for Carbon-Related Environmental Analysis

MSc Research Project · Teluk Panggang, Banyuwangi

- Conducted non-destructive field sampling of mangrove biomass, including DBH, canopy height, species identification, and GNSS-based plot positioning.
- Supported aboveground biomass estimation and mapping using Sentinel-1 and Sentinel-2, including analysis of relationships between biomass and vegetation parameters such as canopy density and LAI.
- Developed practical understanding of the relationship between vegetation condition, biomass, carbon storage, and spatial environmental monitoring.

MSC THESIS

River Discharge Estimation Using Sentinel-2 and the CMHC Method

Master of Remote Sensing · Universitas Gadjah Mada | 2024–2026

- Evaluated the Calibration Measurement Hierarchical Classification (CMHC) method for satellite-based river discharge estimation across the Progo, Oyo, and Serang Rivers.
- Developed an end-to-end workflow using Google Earth Engine and Python, including Sentinel-2 preprocessing, cloud masking, ROI extraction, spectral feature generation, classification, statistical analysis, and model validation.
- Applied Random Forest for discharge-class classification and evaluated model performance using NSE, RMSE, and rRMSE on independent 2022–2024 test data.
- Found that river width and spatial representation of the water body at Sentinel-2 resolution were more influential on model performance than seasonal separation, highlighting the importance of spatial characteristics when applying remote-sensing-based environmental models.

CAPACITY BUILDING & KNOWLEDGE SHARING

Invited Speaker —EAGE Online Training (GE-ONIT)

Remote Sensing for Environmental Monitoring · 2026

- Delivered technical training covering remote sensing fundamentals, image interpretation, land cover/land use classification, environmental monitoring, and Google Earth Engine workflows.
- Demonstrated an end-to-end Sentinel-2 and Random Forest workflow, connecting data → analysis → environmental information → decision support.
- Developed and shared a documented, reproducible workflow for practical participant use.

EDUCATION

Master of Remote Sensing

2024–2026

Universitas Gadjah Mada · GPA: 3.89 / 4.00

Thesis: Comparative Study of River Discharge Estimation Performance Using the Calibration Measurement Hierarchical Classification (CMHC) Method Based on Sentinel-2 Imagery

Bachelor of Geography

2017–2021

Universitas Negeri Padang · GPA: 3.81 / 4.00

Undergraduate Thesis: Land Cover Prediction and Surface Runoff Discharge Estimation in the Batanghari Hilir Sub-Watershed for 2030

PROFESSIONAL CERTIFICATION

- Certified GIS Information System Specialist — BNSP, Level 7 (2026–2029)
- Certified GIS Information System Analyst — BNSP, Level 6 (2022–2025)

TECHNICAL SKILLS

GIS & Spatial

QGIS · ArcGIS Pro · WebGIS · spatial overlay · spatial analysis · watershed analysis · multi-criteria analysis

Remote Sensing

spectral indices · image classification · environmental monitoring

Programming & Data

Python · NumPy · Pandas · scikit-learn · PyQGIS · Google Colab

Environmental Modelling

QSWAT+ · HEC-HMS · HEC-RAS · hydrological analysis

Data & Methods

Random Forest · Pearson correlation · regression · R^2 · NSE · RMSE · statistical validation

SELECTED PUBLICATIONS

- **Influence of ENSO on Rainfall Variability in Central Sulawesi** — Buletin GAW Bariri, 2025
- **Flood Potential Assessment of the Way Urang Sub-Watershed Based on Peak Discharge Using the Rational Method** — International Journal for Disaster and Development Interface, 2025
- **Application of Landsat-8 Imagery for Total Suspended Solid (TSS) Mapping in Lambeuso Bay** — Jurnal Spasial, 2019
- **Rainfall Correlation with Tidal Energy Potential in Bungus Bay** — Proceedings of the National Geography Seminar, 2019